

MA416

Lecture 02 – September 4, 2025

Review on Two-Sample Means & FPC's

Dr. Matthew N. Moore

Common Terminology

We start with two populations, P_1 and P_2, whose associated means are μ_1 and μ_2 and variances σ_1^2 and σ_2^2, respectively. For the entirety of this set of notes, we will assume that P_1 and P_2 are **independent**.

Terminology:

Definition 1: Two populations are homoscedastic to one another if $\sigma_1^2 = \sigma_2^2$. If they are not homoscedastic, we will say that they are heteroscedastic.

Definition 2: Two populations are homogeneous to one another if $\mu_1 = \mu_2$. If they are not homogeneous, we will say that they are heterogeneous.

Base strategy for estimation (you should know why…), for $k = 1,2$:

- We will use $\bar{x}_k$ to approximate μ_k
- We will use s_k^2 to approximate σ_k^2

Test for Homoscedasticity (2 independent samples)

We start by assessing the homoscedasticity (or lack thereof) between two independent populations. Suppose you gather a simple random sample from each of the independent populations.

Sample 1: Sample size = n_1, sample variance = s_1^2
Sample 2: Sample size = n_2, sample variance = s_2^2

The null/alternative pair we will default to will be:

$$H_0: \sigma_1^2 = \sigma_2^2$$
$$H_a: \sigma_1^2 \neq \sigma_2^2$$

Equivalent null hypotheses:

$$H_0: \sigma_d^2 = 0 \quad \text{where} \quad \sigma_d^2 = \sigma_1^2 - \sigma_2^2$$

$$H_0: \sigma_q^2 = 1 \quad \text{where} \quad \sigma_q^2 = \frac{\sigma_1^2}{\sigma_2^2}$$

Test for Homoscedasticity (2 independent samples)

General preferred choice:

$$H_0: \sigma_q^2 = 1 \quad \text{where} \quad \sigma_q^2 = \frac{\sigma_1^2}{\sigma_2^2}$$

Question: Why?

We plan to use S_k^2 to estimate σ_k^2, implying we will use S_q^2 to estimate σ_q^2; and since S_1^2 and S_2^2 are independent $\chi^2_{n_k-1}$ random variables (from probability theory, under the assumption of independence AND normality), then:

$$S_q^2 = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1, n_2-1)}$$

We will then choose:

$$F_{stat} \coloneqq \frac{S_1^2}{S_2^2}$$

with the chosen preference that $S_1^2 \geq S_2^2$, giving us $F_{stat} \geq 1$

Test for Homoscedasticity (2 independent samples)

$$F_{stat} := \frac{S_1^2}{S_2^2} \geq_c 1$$

In terms of design, this means that if $F_{stat} \gg 1$, then $S_1^2 \gg S_2^2$, implying that σ_1^2 is possible significantly larger than σ_2^2, implying that it is likely that $\sigma_1^2 \neq \sigma_2^2$, implying that the populations are heteroscedastic populations.

Hence our decision rule becomes:

- If $F_{stat} \gg 1$, we reject H_0; implying that we have evidence that P_1 and P_2 are heteroscedastic
- If $F_{stat} \approx 1$, we fail to reject H_0; implying that we have evidence that P_1 and P_2 are homoscedastic

Test for Homoscedasticity (2 independent samples)

```
S1 = c(34, 72, 45, 89)
S2 = c(47, 83, 41, 39, 89, 74)
c(sd(S1), sd(S2))
```

```
> c(sd(S1), sd(S2))
[1] 25.07323 22.40015
```

```
n1 = length(S1)
n2 = length(S2)

# caution on the next line!!!
Fstat = var(S1)/var(S2)
Fstat
```

```
> Fstat
[1] 1.252906
```

```
alpha = 0.05
fcrit = qf(alpha, n1-1, n2-1, lower.tail=F)
pval = pf(Fstat, n1-1, n2-1, lower.tail=F)

c(fcrit, pval)
```

```
> c(fcrit, pval)
[1] 5.4094513 0.3840727
```

Test for Homogeneity (2 independent samples)

The null/alternative pair we will default to will be:

$$H_0: \mu_1 = \mu_1$$
$$H_a: \mu_1 \neq \mu_2$$

Equivalent null hypothesis:

$$H_0: \mu_d = 0 \quad \text{where} \quad \mu_d = \mu_1 - \mu_2$$

We will estimate μ_d with $\bar{x}_d$, where $\bar{x}_d = \bar{x}_1 - \bar{x}_2$.

Question 1: Unbiased?

$$\begin{aligned}\mathbb{E}(\bar{X}_d) &= \mathbb{E}(\bar{X}_1 - \bar{X}_2) \\ &= \mathbb{E}(\bar{X}_1) - \mathbb{E}(\bar{X}_2) \\ &= \mu_1 - \mu_2 \\ &= \mu_d\end{aligned}$$

Test for Homogeneity (2 independent samples)

Question 2: Consistent?

Note: Since $\bar{X}_d$ is unbiased for μ_d, we only need to check to see if its variance tends to 0 for large n_1, n_2.

$$\begin{aligned}\mathbb{V}(\bar{X}_d) &= \mathbb{V}([1 \cdot \bar{X}_1] + [-1 \cdot \bar{X}_2]) \\ &= (1)^2 \cdot \mathbb{V}(\bar{X}_1) + (-1)^2 \cdot \mathbb{V}(\bar{X}_2) \cancel{+ 2 \cdot (1) \cdot (-1) \cdot Cov(\bar{X}_1, \bar{X}_2)} \\ &= \mathbb{V}(\bar{X}_1) + \mathbb{V}(\bar{X}_2) \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\end{aligned}$$

Therefore:

$$\mathrm{SE}(\bar{X}_d) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \rightarrow^{(n_1 \to \infty \ \& \ n_2 \to \infty)} \rightarrow 0$$

Test for Homogeneity (2 independent samples)

Test Statistic:

$$\star_{stat} = \frac{\bar{X}_d - \mu_d^{(0)}}{\mathrm{SE}(\bar{X}_d)}$$

If H_0 is true (i.e. $\mu_d = \mu_d^{(0)} =^{\text{typically}} = 0$), then $\star_{stat} \sim Z$, provided σ_1^2, σ_2^2 are both known.

Case 1: Assume that P_1 is homoscedastic to P_2; i.e. $\sigma_1^2 = \sigma_2^2 := \sigma^2$

Therefore:

$$\mathrm{SE}(\bar{X}_d) = \sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}} = \sigma \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Hence, we just need to approximate σ^2.

Test for Homogeneity (2 independent samples)

Theorem: Define $S_p^2 := \frac{S_1^2(n_1-1)+S_2^2(n_2-1)}{n_1+n_2-2}$. Then S_p^2 is both an unbiased and consistent estimator for σ^2.

Then:

$$\mathrm{SE}(\bar{X}_d) \approx \sqrt{\frac{S_1^2(n_1-1)+S_2^2(n_2-1)}{n_1+n_2-2}} \cdot \sqrt{\frac{1}{n_1}+\frac{1}{n_2}} = S_p \cdot \sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$$

In this case, one can show:

$$\star_{stat} = \frac{\bar{X}_d - \mu_d^{(0)}}{\mathrm{SE}(\bar{X}_d)} \sim T_\nu, \qquad \nu = n_1 + n_2 - 2$$

Test for Homogeneity (2 independent samples)

Case 2: Assume that P_1 and P_2 are heteroscedastic; i.e. $\sigma_1^2 \neq \sigma_2^2$

Therefore:

$$\mathrm{SE}(\bar{X}_d) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \approx \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

In this case, one can show:

$$\star_{stat} = \frac{\bar{\boldsymbol{X}}_{\boldsymbol{d}} - \mu_d^{(0)}}{\mathrm{SE}(\bar{X}_d)} \sim T_\nu$$

where

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2(n_1 - 1)} + \frac{s_2^4}{n_2^2(n_2 - 1)}}$$

Test for Homogeneity (2 independent samples) Summary

Question 1: Are both populations normally distributed?
- If no, use non-parametric methods
- If yes, go to question 2
- If you don't know, assess (methods to explore later)

Question 2: Do you know both population variances?
- If yes, use Z
- If no, use T

Question 3: Are the populations homoscedastic?
- If yes, use the pooled-variance T-test
- If no, use the unpooled-variance T-test
- If you don't know, use the F-test for homoscedasticity to assess.

Finite Population Acknowledgement

Recall that:

- n represents sample size
- N represents population size, with $n \leq N$

Therefore, if $n = N$, then:

- $\bar{x} = \mu$ (mean)
- $s_x^2 = \sigma^2$ (variance)
- $r = \rho$ (correlation)
- $s_{x,y} = \sigma_{x,y}$ (covariance)

Hence in theory, it **should** be the case that:

$$\mathrm{SE}(\bar{X}) = \mathrm{SE}(S^2) = \mathrm{SE}(r) = \cdots = 0$$

but...

$$\mathbb{V}(\bar{X}) = \frac{\sigma^2}{n} \to \frac{\sigma^2}{N} \neq 0 \quad \text{if} \quad N \in \mathbb{N}$$

Finite Population Correction Factor

Theorem: If N is known and is finite, then:

$$\mathbb{V}(\bar{X}) = \frac{\sigma^2}{n} \cdot \left(\frac{N-n}{N-1}\right)$$

where $\frac{N-n}{N-1}$ is known as the "finite population correction" (fpc) factor.

Important properties:

- If $n = 1$, $f_{pc} = 1$
- If $n = N$, $f_{pc} = 0$
- For any $n \in \{1, 2, \ldots, N\}$, $f_{pc} \in [0, 1]$
- If $f(n) = \frac{N-n}{N-1}$, then $\frac{df}{dn} < 0$ (that is, if you increase n, f_{pc} decreases)

Full Example

```
# Suppose you choose:
# Homoscedasticity alpha = 15%
# Homogeneity alpha = 8%
# Suppose you know P1 and P2 are normally distributed

S1 = c(23, 21, 25, 26, 29, 19, 21)
S2 = c(23, 21, 22, 21, 19, 25, 23, 21, 23, 21)

xbar1 = mean(S1)
xbar2 = mean(S2)
xbard = xbar1 - xbar2

sx1 = sd(S1)
sx2 = sd(S2)
fstat = sx1^2/sx2^2

n1 = length(S1)
n2 = length(S2)

c(xbar1, xbar2, xbard)
c(sx1, sx2, fstat)
```

```
> c(xbar1, xbar2, xbard)
[1] 23.428571 21.900000  1.528571
> c(sx1, sx2, fstat)
[1] 3.457222 1.663330 4.320138
```

Full Example

```
alpha_var = 0.15 # chosen
pf(fstat, n1-1, n2-1, lower.tail=F)
```

Since $p_{val}^{\sigma_1^2=\sigma_2^2} < \alpha^{\sigma_1^2=\sigma_2^2}$, we have evidence to believe that the variances are not equal; hence we proceed with an unpooled-variance T-test structure.

```
sexbard = sqrt(sx1^2/n1 + sx2^2/n2)
tstat = (xbard - 0)/sexbardtstat
```

```
vtop = (sx1^2/n1 + sx2^2/n2)^2
vbot = sx1^4/(n1^2*(n1-1)) + sx2^4/(n2^2*(n2-1))
v = vtop/vbot
c(n1-1, n2-1, v)
```

```
pval = pt(abs(tstat), v, lower.tail=F)*2
pval
```

```
> c(xbar1, xbar2, xbard)
[1] 23.428571 21.900000  1.528571
> c(sx1, sx2, fstat)
[1] 3.457222 1.663330 4.320138
```

```
> pf(fstat, n1-1, n2-1, lower.tail=F)
[1] 0.02499291
```

```
> tstat
[1] 1.085172
```

```
> c(n1-1, n2-1, v)
[1] 6.000000 9.000000 7.962541
```

```
> pval
[1] 0.3096044
```

Be able to interpret correctly!